SnpEff: Variant analysis

Contents

Summary
Variant rate by chromosome
Variants by type
Number of variants by impact
Number of variants by functional class
Number of variants by annotation
Quality histogram
InDel length histogram
Base variant table
Transition vs transversions (ts/tv)
Allele frequency
Allele Count
Codon change table
Amino acid change table
Chromosome variants plots
Details by gene

Summary

Genome	reference_db
Date	2025-12-07 16:43
SnpEff version	`SnpEff 5.4a (build 2025-11-25 12:22), by Pablo Cingolani`
Command line arguments	`SnpEff -stats results/snpEff/snpEff.html reference_db results/variants/filtered_variants.vcf`
Warnings	0
Errors	0
Number of lines (input file)	19
Number of variants (before filter)	21
Number of non-variants (i.e. reference equals alternative)	0
Number of variants processed (i.e. after filter and non-variants)	21
Number of known variants (i.e. non-empty ID)	0 (0%)
Number of multi-allelic VCF entries (i.e. more than two alleles)	2
Number of annotations	42
Genome total length	15,350
Genome effective length	15,350
Variant rate	1 variant every 730 bases

Variants rate details

Chromosome	Length	Variants	Variants rate
NG_011815.1	15,350	21	730
Total	**15,350**	**21**	**730**

Number variants by type

Type	Total
SNP	19
MNP	0
INS	2
DEL	0
MIXED	0
INV	0
DUP	0
CNV	0
BND	0
INTERVAL	0
Total	**21**

Number of effects by impact

Type (alphabetical order)	Count	Percent
MODIFIER	42	100%

Number of effects by functional class

Type (alphabetical order)	Count	Percent

Missense / Silent ratio: 0

Number of annotaitons and region counts

Annotation

Type (alphabetical order)	Count	Percent
intergenic_region	21	50%
upstream_gene_variant	21	50%

Region

Type (alphabetical order)	Count	Percent
INTERGENIC	21	50%
UPSTREAM	21	50%

Quality:

Min	35
Max	584
Mean	230.895
Median	134
Standard deviation	197.109
Values	35,46,61,69,71,88,98,134,137,153,187,257,346,497,510,516,527,584
Count	1,1,1,1,2,1,1,1,1,1,1,1,1,1,1,1,1,1

Insertions and deletions length:

Min	0
Max	0
Mean	0
Median	0
Standard deviation	0
Values	0
Count	2

Base changes (SNPs)

	A	C	G	T
A	0	3	5	0
C	0	0	0	3
G	0	0	0	0
T	2	5	1	0

Ts/Tv (transitions / transversions)

Note: Only SNPs are used for this statistic.
Note: This Ts/Tv ratio is a 'raw' ratio (ratio of observed events).

Transitions	17
Transversions	6
Ts/Tv ratio	2.8333

All variants:

```
Sample ,sample1,Total
Transitions ,17,17
Transversions ,6,6
Ts/Tv ,2.833,2.833
```

Only known variants (i.e. the ones having a non-empty ID field):

```
No results available (empty input?)
```

Allele frequency

Min	50
Max	100
Mean	76.316
Median	50
Standard deviation	25.649
Values	50,100
Count	9,10

Allele Count

Min	1
Max	2
Mean	1.526
Median	1
Standard deviation	0.513
Values	1,2
Count	9,10

Hom/Het per sample

```
Sample_names , sample1
Reference , 0
Het , 9
Hom , 10
Missing , 0
```

Codon changes

How to read this table:
- Rows are reference codons and columns are changed codons. E.g. Row 'AAA' column 'TAA' indicates how many 'AAA' codons have been replaced by 'TAA' codons.
- Red background colors indicate that more changes happened (heat-map).
- Diagonals are indicated using grey background color
- WARNING: This table may include different translation codon tables (e.g. mamalian DNA and mitochondrial DNA).

Amino acid changes

How to read this table:
- Rows are reference amino acids and columns are changed amino acids. E.g. Row 'A' column 'E' indicates how many 'A' amino acids have been replaced by 'E' amino acids.
- Red background colors indicate that more changes happened (heat-map).
- Diagonals are indicated using grey background color
- WARNING: This table may include different translation codon tables (e.g. mamalian DNA and mitochondrial DNA).

Variants by chromosome

```
NG_011815.1, Position,0,100,200,300,400,500,600,700,800,900,1000,1100,1200,1300,1400,1500,1600,1700,1800,1900,2000,2100,2200,2300,2400,2500,2600,2700,2800,290
NG_011815.1,Count,7,14,0,0,0
```

Details by gene

Here you can find a tab-separated table.